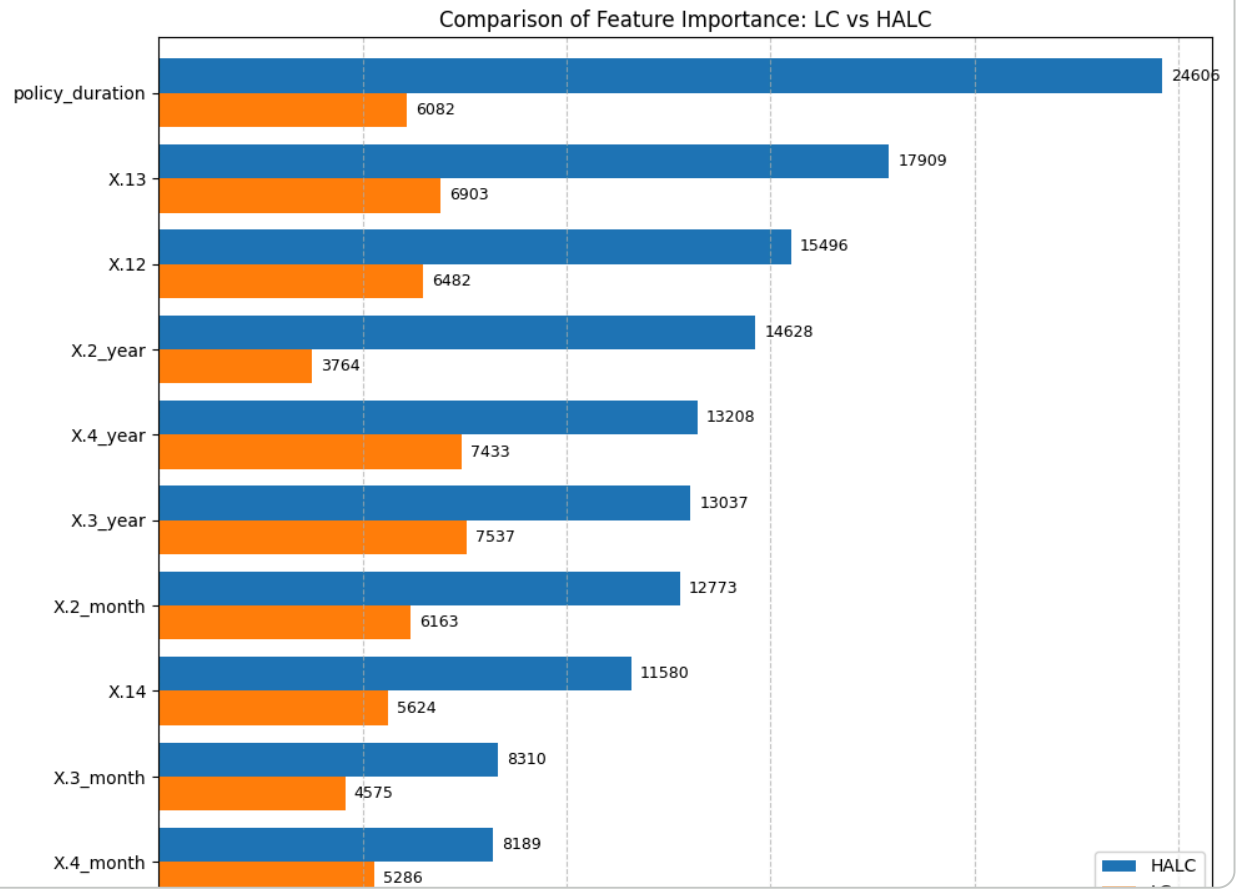
```
# 5. Rounding the labels (the fix from before)
for p in ax.patches:
    val = p.get_width()
    ax.annotate(f'{val:.0f}',
                (val, p.get_y() + p.get_height() / 2),
                ha='left', va='center', xytext=(5, 0),
                textcoords='offset points', fontsize=9)

plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



CS Model Construction and Prediction

Below, we build a classification model for `CS`, evaluate it on a validation split, and then generate predicted claim probabilities for the test set.

```
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import roc_auc_score, classification_report, confusion_matrix

# # Build a modeling copy so the earlier cleaned dataframes stay unchanged
```